

EduGuard

Hybrid Guardrail System for Educational Support Chatbots

Sultanah Alharbi, Budor Alharbi

Jun 2026

Project Report

1. Project Overview

EduGuard is an AI system that helps keep university students safe from cheating, mental health issues, and harmful messages sent to chatbots. It uses a mix of different tools like rule-based checks, a special kind of AI model called DistilBERT, and a system called FAISS to help find and use relevant information. These tools work together to make sure student questions are safe and appropriate before they reach the AI chatbot.

EduGuard is built using Python, PyTorch, HuggingFace Transformers, and Streamlit. It runs in real time, checking every message a student sends before it goes to the language model. This ensures only safe and relevant questions get a response from the AI, which helps reduce risk and saves on API costs.

1. What We Built

EduGuard is a mix of AI safety tools that help protect an educational chatbot from unsafe or harmful messages from students.

Instead of sending every message directly to GPT, EduGuard first checks the message with:

- Rule-based filtering
- A fine-tuned DistilBERT classifier
- A decision layer
- FAISS-based RAG retrieval
- GPT response generation (SAFE only)

Only safe academic questions reach GPT, lowering the risk and cost of using the API.

2. Who It Is For

EduGuard serves two audiences:

- Students — through a Streamlit chat interface
- University administrators, who use an admin dashboard to see flagged messages, how serious they are, and whether they've been reviewed

It is designed for universities deploying AI chatbots with built-in safety.

3. How It Works

Every message passes through five stages:

Stage	Component	What It Does
1	Rule-Based Filter	Catches obvious unsafe patterns
2	DistilBERT Classifier	Assigns one of five labels
3	Decision Layer	Routes the message
4	FAISS RAG	Retrieves relevant academic context
5	GPT Response	Generates final answer (SAFE only)

Safety Labels

Label	Meaning	Action
SAFE	Academic question	Answer using RAG + GPT
UNSAFE	Cheating request	Refuse + redirect
SENSITIVE	Wellbeing concern	Support + escalate
OUT_OF_SCOPE	Not academic	Redirect
PROMPT_INJECTION	Bypass attempt	Block + alert

3. AI Components Used

- DistilBERT Classifier
 - Fine-tuned on 800 synthetic samples
 - 160 samples per label
 - 2 epochs
- FAISS RAG
 - Embedding model: all-mpnet-base-v2
 - Retrieves top-3 chunks
 - Threshold: 0.45
- GPT Layer

- Generates final answers for SAFE messages only
- Rule-Based Filter
- Zero-cost keyword matching
- Catches cheating, crisis terms, and injection attempt

4. What Worked Well

- The system had an accuracy of 95% in testing
- No unsafe messages were incorrectly marked as safe
- The combination of tools kept the system fast and efficient
- The admin dashboard clearly showed different levels of severity
- The RAG system was reliable and consistent
- The mock mode let testers check the system without using real API calls

5. What Did Not Work / Incomplete

- Five safe messages were incorrectly marked as unsafe
- The rule-based filter didn't catch some disguised cheating attempts
- The knowledge base used was made up, not real data
- There was no way to track conversation history
- The admin dashboard didn't always update properly in the background

6. What We Would Improve

- Include real university policy documents
- Use real, anonymized student data instead of fake data
- Add the ability to understand conversations over multiple turns
- Add a way to track and resolve alerts, plus send notifications
- Test the system against harmful or misleading messages through red-team methods
- Explore using even more powerful transformer models

Summary Table

Metric	Value
Training dataset	800 samples
Benchmark dataset	100 samples
Benchmark accuracy	95%
Benchmark errors	5 (all SAFE misclassified)
Classifier model	DistilBERT
Embedding model	all-mpnet-base-v2

RAG threshold	0.45
GPT usage	SAFE messages only

Final Summary

EduGuard demonstrates that a hybrid safety pipeline can reliably protect an educational chatbot from cheating, unsafe content, prompt injection, and off-topic misuse — while still providing helpful, grounded academic support.